

Are Archived NBA Consensus Moneylines Calibrated? A Pre-Specified Test of Favorite Pricing Across 15,351 Games

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A pre-specified look at whether favorites in an archived NBA moneyline series were priced fairly, using a verified public archive

Abstract

Tests for bias in sports-betting prices often rely on small groups of games. That creates a power problem: a significant result can be fragile, while a non-significant result may simply be too weak to detect a meaningful effect. This study set its analysis strategy before looking at outcomes, defined the smallest effect worth detecting as the margin a bettor would need to overcome at the quoted prices, and then ran one primary test on 15,351 NBA regular-season games from 2007-08 through 2019-20.

Favorites with a vig-free implied probability above .70 (6,840 games) won 80.83 percent of the time, compared with an implied 80.26 percent. That is a 0.58 percentage-point gap ($z = 1.22$, $p = .223$). The estimate is stable across six variance estimators that allow correlation within seasons and within teams, none of which rejects calibration. At the quoted prices a bettor needed favorites to outperform their vig-free probability by 3.01 percentage points to break even, against a design able to detect 1.33 points, so the observed gap is well inside the region the test was built to rule out.

That conclusion is not universal, and the limits are stated here rather than left to the appendix. Repeating the test under five vig-removal rules gives gaps from -1.31 to +0.58 points. Four of the five leave the gap below that rule's own break-even threshold and none of those four is significant. The fifth, power normalization, produces a gap of -1.31 points against a 1.24-point threshold at $p = .004$. Separately, the secondary logistic calibration regression does not reject perfect calibration under four of five variance estimators but does under two-way clustering on season and team ($p = .037$), an estimator resting on only 13 clusters and therefore unreliable.

Taken together the more durable finding is methodological. On the same 15,351 games, raw implied probabilities imply a favorite bias of -2.50 points at $p < .001$, proportional normalization implies +0.58 points at $p = .223$, and power normalization implies -1.31 points at $p = .004$. The existence, size and even the sign of estimated favorite bias in this market depend materially on how the bookmaker's margin is removed, and a pre-specified design, adequately powered for its primary estimand and anchored to an economic threshold, is what makes that dependence visible instead of invisible.

Keywords: sports betting, market calibration, favorite-longshot bias, pre-specified analysis, vig removal, statistical power, NBA

1. Introduction

Legal sports betting is now a large consumer market in the United States. State-regulated sportsbooks accepted roughly \$149.9 billion in wagers during 2024 (American Gaming Association, 2025). That scale makes it worth asking how well prices work, but it does not make every statistical finding about those prices convincing. A dataset can include thousands of games and still leave only a few dozen observations in the odds range that builds a conclusion.

A short power calculation makes the issue clear. Take a 74-game group of favorites priced near .85, the kind of narrow bucket often used in these studies, and ask whether it could detect a deviation the size of the break-even margin computed later from these very prices, 3.01 percentage points. It has about 11 percent power to do so. Even samples of several hundred games often miss moderate deviations. That creates two risks: a significant result from a small group can turn on one or two outcomes, and a null result can be uninformative because the test was never able to detect the effect in the first place.

This study is built to avoid that trap, and it asks one question:

Do the archived consensus moneylines in this series systematically misstate the win probability of favorites by an amount large enough to be both statistically detectable and economically exploitable?

The population is NBA regular-season games from 2007-08 through 2019-20. The primary estimand is the calibration gap among favorites whose vig-free implied probability exceeds .70. The threshold, the analysis plan, and the smallest effect worth detecting were all fixed before outcomes were examined. With 6,840 games in that group the design detects a deviation of 1.33 percentage points at 80 percent

power, comfortably below the 3.01-point break-even requirement a bettor actually faces. Whatever the outcome, the result is informative.

1.1 Three questions that are easy to conflate

This literature moves quickly between three claims that are related but not equivalent, and this paper is careful about which one it makes.

Calibration asks whether stated probabilities match realized frequencies. Profitability asks whether a betting rule earns a positive return after paying the quoted price. Market efficiency asks whether available information is impounded in prices so thoroughly that no systematic strategy can exploit it.

A market can be well calibrated on average and still contain predictable conditional patterns. It can also carry small calibration errors that no bettor could monetize once the margin is paid. This paper's primary estimand is calibration, with a secondary and direct test of flat-stake profitability. It is not a test of general informational efficiency, and no such claim is made.

2. Background

Fama (1970) frames market efficiency as a question about what information prices absorb. Betting markets offer a clean version of that test because every price settles against a known outcome. Early point-spread work on professional football generally found that simple public betting rules did not produce stable profits (Zuber, Gandar, & Bowers, 1985; Sauer, Brajer, Ferris, & Marr, 1988), although profitability, calibration, and price efficiency are connected but distinct questions.

Four strands of that literature bear on this study, and they disagree with one another in ways worth naming.

The first is behavioral. Longshot bias means longshots are overpriced and return less than favorites, the classic result in horse racing (Griffith, 1949; Thaler & Ziemba, 1988). Favorite bias is the reverse. Because the labels are used inconsistently, this paper follows Newall and Cortis (2021). Woodland and Woodland (2001) report favorite bias in NHL money lines.

The second concerns how bookmakers set prices. Levitt (2004) argues that a book can shade prices toward bettor demand rather than balancing every position. That is a candidate mechanism for mispricing, not evidence that mispricing exists. Paul and Weinbach (2008) find that NBA favorites attract a disproportionate share of tickets but do not find that betting against them beats the market.

The third asks whether any of it survives costs. Moskowitz (2021) finds that behavioral patterns in betting prices are too small to clear transaction costs. Angelini and De Angelis (2019) test odds across many books and find most markets efficient at the best available price.

The fourth is methodological, and it is where this paper sits. Winkelmann, Otting, Deutscher, and Makarewicz (2024) benchmark inefficiency tests against simulated and real odds and ask whether the tests themselves are up to the job. What is still missing is an NBA moneyline calibration analysis with a pre-specified comparison and an explicit power justification. This paper is not trying to rediscover favorite bias. It asks whether a test adequately powered for a pre-specified estimand, and anchored to an economic threshold, can separate real favorite mispricing from the artifact created by the bookmaker's margin.

3. Data

The data are consensus moneylines and final scores for 15,490 NBA regular-season games from 2007-08 through 2019-20, originally published by the sportsbookreviewsonline.com archive and redistributed in the public repository accompanying Dotan's UCLA master's thesis (Dotan, 2020). Files were retrieved unmodified on 20 July 2026.

The archive reports one moneyline for each side of a game, but it does not say whether the line is an opening or closing quote, and it does not identify the originating sportsbook. This paper therefore calls the object an archived consensus moneyline and avoids treating it as a closing line. That distinction is not cosmetic: a price recorded in the morning and a price recorded seconds before tip-off are different economic objects, and nothing in this dataset distinguishes them. One further sign that the series is a composite rather than a single book's quote is that one game is recorded with a booksum below one, which no single book would offer.

3.1 Verification

Because the data passed through two intermediaries, they were audited before analysis began. A stratified sample of 260 games, 20 from each of the 13 seasons, was checked directly against the live primary archive. All 260 moneylines matched exactly on both sides, including extreme prices such as -1300 against +800, and all 260 outcomes matched the final scores.

That sample was drawn by hand rather than by a seeded random procedure, so it should be read as a stratified spot check that found no errors, not as a probabilistic

guarantee about the remaining games. The stronger claim rests on an automated check that covers the whole file. The same source repository ships a separate team-game-level extraction with its own moneyline and outcome columns, and every value in the analysis file is compared against it: 30,978 prices and 15,490 outcomes, with no discrepancy. Season game counts match the true NBA schedule in every year, including the 990-game 2011-12 lockout season, the 1,229-game 2012-13 season after one cancellation, and the 971 games played before the March 2020 suspension. One game is missing a moneyline in the source, matching the source thesis's own description. The verification script and its full output are published with the replication materials.

3.2 Exclusions and profile

The analysis excludes one game with a missing moneyline and 138 pick'em games, where neither side is a favorite. That leaves 15,351 games. The source thesis's processing pipeline had already removed 986 playoff games, so the analysis covers the regular season only. The mean overround on the analysis sample is 3.77 percent, and favorite vig-free implied probabilities range from .502 to .985.

The series ends in March 2020 because that is where the source archive ends. It therefore predates the subsequent expansion of legal sports betting in the United States. No claim is made here about whether that expansion changed market behavior; it is simply outside the sample.

4. Methods

The analysis plan was written and frozen before any outcome data were examined, and it is published in full with the replication materials. The record of that freeze is internal to the author's files rather than lodged with an outside registry, so the work is described throughout as pre-specified rather than pre-registered. The limitations section explains why that distinction matters.

4.1 Probabilities and the favorite

American odds convert to a raw implied probability. For negative odds q equals the absolute value of the odds divided by that value plus 100; for positive odds q equals 100 divided by the odds plus 100. The two raw probabilities of a game sum to more than one, and the excess is the bookmaker's margin.

Writing the two raw probabilities as q_1 and q_2 , and their sum as P , the primary analysis removes the margin proportionally:

$$p1 = q1 / P \quad \text{and} \quad p2 = q2 / P$$

This assumes the margin is taken in proportion to price, so that in absolute terms the favorite absorbs more of it than the underdog. The rule has one clean property: dividing both sides by the same constant leaves the ratio $q1/q2$ unchanged, so the relative pricing of the two sides is preserved exactly and only the level is rescaled. The additive rule preserves the difference $q1 - q2$ instead. Neither is derivable from first principles; each preserves a different feature of the quoted price.

Three things should be said plainly about why proportional is the primary rule, because the choice determines the headline number. First, it was fixed in the frozen analysis plan before any outcome data were examined, and that plan is published with the replication materials, so it was not selected after seeing which rule produced which answer. Second, it is the prevailing convention in the applied betting-market literature, which makes the primary estimate comparable with prior work. Third, its property of preserving relative pricing is the natural default for a calibration question, where the object of interest is whether the two sides are priced correctly against each other.

What should not be claimed is that proportional normalization has stronger theoretical warrant than the alternatives. It does not. Of the rules tested, the one with an actual behavioral model behind it is Shin (1993), and for a two-outcome book Shin reduces exactly to the additive rule, which puts the gap at -0.60 rather than +0.58 points. That estimate is not significant ($p = .186$) and sits well below its own break-even requirement, so the substantive conclusion is unchanged, but the sign is not. Section 5.4 reports all five rules and Appendix A proves the Shin result. The favorite is the side with the higher vig-free probability. Raw probabilities are retained for the sensitivity analysis.

4.2 The primary test

The single confirmatory hypothesis concerns favorites whose vig-free implied probability exceeds .70. Under the null each game is a Bernoulli trial with its own probability, so the number of favorite wins follows a Poisson-binomial distribution with mean equal to the sum of the game-level probabilities and variance equal to the sum of $p(1 - p)$. The test compares actual wins with that expected sum and uses the corresponding game-level null variance. It is two-sided at $\alpha = .05$. Because implied probabilities vary game by game, this construction replaces the fixed-probability bucket tests common in the literature, which treat every game in a bucket as carrying the same probability.

4.3 Dependence

The Poisson-binomial framework treats games as independent conditional on their own prices. That is standard for a calibration test, but these observations are not obviously independent, and the assumption deserves to be named rather than assumed.

Dependence within a game is not an issue, because only one favorite outcome is analyzed per game and the two sides are collinear by construction. Four other channels are plausible. Each team appears in roughly a thousand games, so a team the market persistently misjudges would correlate its own residuals across the sample. Roster rules, pace, officiating and market structure are common within a season. Injuries and news propagate across nearby games. And a consensus price aggregates books that share information, so pricing errors need not be idiosyncratic.

None of this changes the point estimate; all of it can change the standard error. Section 5.4 therefore re-estimates the gap under cluster-robust variance at the season level, the favorite-team level, the underdog-team level, and two-way, and adds a team-blocked bootstrap alongside the season-blocked one. The confirmatory test is retained in its pre-specified form; the question asked of these alternatives is only whether the conclusion survives them.

4.4 Power

The null result matters only if the design could have caught a deviation worth caring about, so the calculation is stated in full. Let p be the vig-free implied probability of each game in the primary group and n the number of games. The standard error of the gap is the square root of the sum of $p(1 - p)$, divided by n . For a two-sided test at α , power against a true gap is the probability that the standardized estimate exceeds the critical value in either tail. Under the normal approximation, the conventional two-sided minimum detectable effect at 80 percent power is the sum of the two critical values multiplied by that standard error. The variance is the heterogeneous per-game one rather than a single pooled value, so the calculation reflects the actual distribution of prices.

Parameter	Value
Primary group n	6,840
Significance level (two-sided)	$\alpha = .05$
Target power	.80
Null model	Poisson-binomial, per-game variance

Parameter	Value
Standard error of the gap	0.474 pp
Critical values $z(\alpha/2)$, $z(\text{power})$	1.960, 0.842
Minimum detectable effect, normal approximation	1.3281 pp
Minimum detectable effect, exact Poisson-binomial	1.3242 pp
Break-even requirement	3.01 pp
Observed gap	+0.58 pp

Table 1. Power parameters for the primary test. The normal-approximation MDE is the sum of the two critical values multiplied by the standard error; the exact figure inverts the Poisson-binomial power function directly. Figure 3 plots the full power curve behind these numbers.

That formula is an approximation, not an exact inversion of the Poisson-binomial power function, so the exact version was computed as a check. Building the exact Poisson-binomial distribution by discrete Fourier transform of its characteristic function gives an exact two-sided rejection region of 5,425 wins or fewer, or 5,554 or more, with true size .0484 rather than .05 because the statistic is discrete. Inverting exact power at 80 percent gives a minimum detectable effect of 1.3242 points against the normal approximation's 1.3281, a difference of four thousandths of a percentage point. The approximation is reported in the text because it is the conventional quantity; it is not doing any work the exact calculation would undo.

4.5 The economic bar

A calibration gap matters to a bettor only when it is larger than the share of the margin carried by the favorite side of the price. That share equals the raw implied probability minus the vig-free probability, computed game by game. In the primary group it averages 3.01 percentage points, with a 10th-to-90th-percentile range of 2.35 to 3.72. Across all favorites it averages 2.61 points, and in the extreme tail 3.02 points. Because the design's minimum detectable effect is 1.33 points, the test can detect favorite mispricing well below the level a bettor could actually exploit.

4.6 Confirmatory and exploratory analyses

The distinction is kept explicit throughout, because it governs how much weight each result can carry.

Confirmatory, fixed in advance	Exploratory, added at review
Primary Poisson-binomial test above .70	Season-by-season gaps and Cochran Q
Logistic calibration regression	Extreme tail above .90

Confirmatory, fixed in advance	Exploratory, added at review
Five fixed probability buckets, Holm corrected	Season-blocked and team-blocked bootstrap
Flat-stake returns	Cluster-robust variance estimators
Raw-probability sensitivity check	Alternative vig-removal rules

Table 2. Only the left column was pre-specified. Results in the right column are reported as exploratory and are not corrected for multiplicity across the family.

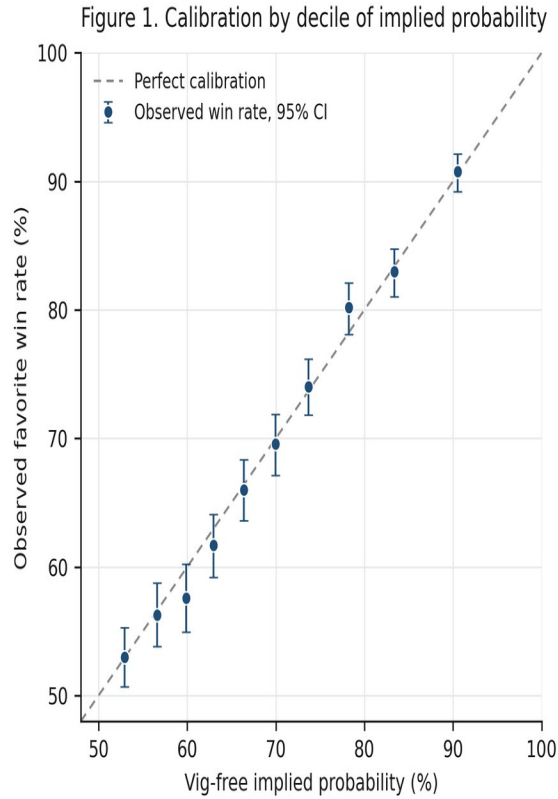
5. Results

5.1 Primary result

The primary group contains 6,840 games, 44.6 percent of the sample. Favorites won 5,529 games against an expected 5,489.45, an observed rate of 80.83 percent against an implied 80.26 percent. The gap is +0.58 percentage points, with $z = 1.22$ and $p = .223$. The test does not reject calibration. The direction of the gap runs toward favorites winning slightly more often than their price implies, which is the opposite of the favorite bias the study was designed to detect.

5.2 Continuous calibration

The logistic regression gives an intercept of -0.047 (95% CI: -0.107 to 0.013) and a slope of 1.049 (95% CI: 0.982 to 1.116). Under classical standard errors the joint Wald test of perfect calibration gives chi-squared = 2.45 with $p = .293$. Section 5.4 reports how that test behaves under clustered variance, where it is less stable than the primary test. Figure 1 shows the decile calibration plot.



Source: consensus NBA moneylines, 2007-08 to 2019-20, 15,351 regular season games. Vig removed proportionally. Deciles of 15,351 games, 1,347 to 1,804 games each.

Figure 1. Observed favorite win rates against vig-free implied probability, by decile of implied probability, with 95 percent Wilson intervals, against the line of perfect calibration. Deciles of 15,351 games, 1,347 to 1,804 games each. Each game is weighted equally. Archived consensus moneylines, 2007-08 to 2019-20, vig removed proportionally.

5.3 Bucket results

None of the pre-specified buckets reject after the Holm correction. The largest deviation is +1.73 points in the .75 to .80 range, with a raw p of .091 and a Holm-adjusted p of .457.

Bucket	n	Observed	Implied	Expected wins	Gap (pp)	z	p	Holm p
(.50, .60]	4,064	54.45%	55.44%	2,253.1	-0.99	-1.27	.204	.817
(.60, .70]	4,447	64.11%	64.85%	2,883.9	-0.74	-1.04	.300	.899
(.70, .75]	1,933	73.10%	72.39%	1,399.3	+0.71	0.70	.485	.970
(.75, .80]	1,661	79.11%	77.38%	1,285.3	+1.73	1.69	.091	.457
(.80, 1.00]	3,246	86.32%	86.41%	2,804.9	-0.09	-0.15	.880	.970

Table 3. Pre-specified probability buckets, Poisson-binomial test in each, Holm correction across the five. Expected wins is the sum of the game-level vig-free probabilities in the bucket.

5.4 Robustness

Three families of check are reported: across seasons, across dependence structures, and across vig-removal rules. The first two leave the primary conclusion intact. The third does not merely qualify it. Under one of five normalization rules the sample shows a statistically significant favorite bias that marginally exceeds its own break-even threshold, which is evidence against universal robustness and is treated here as a finding rather than as a footnote.

Across seasons, no individual season rejects the null. The 13 season-level gaps range from -2.82 to +2.52 percentage points and every p value is above .10. Heterogeneity is assessed with the Cochran Q statistic in its meta-analytic form, which is a test of whether the season gaps differ from one another by more than sampling noise, not a test that the pooled gap is zero. Writing y for the season gap and w for the inverse of its Poisson-binomial null variance, Q is the sum of w times the squared deviation of y from the inverse-variance-weighted mean, referred to a chi-squared distribution on twelve degrees of freedom. Constructed that way, $Q = 8.37$ with $p = .756$ and I -squared at 0 percent, so there is no detectable variation across seasons. In the extreme tail above .90 implied probability, 756 games produce a +0.63-point gap with $p = .506$; its minimum detectable effect of 2.66 points sits just below its 3.02-point break-even requirement, so even the thinnest pre-specified slice is adequately powered for economically relevant bias in that slice.

Dependence is the more serious concern, and the primary estimate is unmoved by it. The most credible single check is the team-blocked bootstrap, because it resamples 32 favorite teams rather than 13 seasons and therefore does not depend on asymptotics that fail at small cluster counts. It gives a 95 percent interval of -0.52 to +1.57 percentage points, wider than the season-blocked interval of -0.19 to +1.26, and containing zero.

Table 4 adds the cluster-robust estimators. The point estimate is +0.58 points in every specification by construction; what changes is the standard error, which ranges from 0.39 to 0.53 points against the Poisson-binomial value of 0.47. No estimator rejects calibration. The two rows resting on 13 season clusters are reported as sensitivity checks and should not be read as reliable finite-sample inference: cluster-robust variance is asymptotic in the number of clusters, and at 13 it is known to understate variance, so the season-clustered p of .134 is if anything too small rather than too large.

Variance estimator	Cluster s	SE (pp)	z	p	95% CI (pp)
Poisson-binomial (primary test)	—	0.47	1.22	.223	-0.35 to +1.51
Independent, classical	—	0.47	1.23	.218	-0.34 to +1.50
Clustered by season (13 clusters, unreliable)	13	0.39	1.50	.134	-0.18 to +1.33
Clustered by favorite team	32	0.53	1.09	.276	-0.46 to +1.62
Clustered by underdog team	33	0.50	1.16	.246	-0.40 to +1.56
Two-way, season and team (13 clusters, unreliable)	13	0.48	1.22	.224	-0.35 to +1.51

Table 4. The primary calibration gap under six variance estimators. The gap is +0.58 pp throughout; only its standard error changes, and none rejects calibration at .05. Estimators built on 13 season clusters are flagged: cluster-robust variance is asymptotic in the cluster count and is not reliable at that number. The team-blocked bootstrap over 32 clusters, 95% CI -0.52 to +1.57 pp, is the more dependable check.

The logistic calibration regression is less stable than the primary test under the same treatment. Its joint Wald p is .293 with classical errors, .221 clustering on favorite team, .181 on underdog team, .085 on season, and .037 two-way on season and team. The last of those would reject perfect calibration at .05. It is reported rather than omitted, with the caveat that cluster-robust variance is unreliable when the number of clusters is small, and thirteen seasons is small enough that the estimator is known to understate variance and overstate significance. The reading this paper takes is that the confirmatory result is robust, the secondary regression is sensitive to how dependence is modeled, and a reader who prefers the two-way estimate should treat the calibration slope as unresolved rather than confirmed.

Vig removal is the assumption the whole analysis rests on, so four alternatives to proportional normalization were run. Under additive normalization each side gives up the same number of probability points. Under power normalization each raw probability is raised to a common exponent chosen so the pair sums to one. Under the constant-odds-ratio rule the quoted and fair odds differ by a common factor. Shin (1993) prices against a share of insider money, which is designed to load more margin onto longshots.

Shin turns out to be redundant here. For a two-outcome book its solution coincides exactly with the additive rule, because the sum constraint together with Shin's difference-preserving property leaves only one admissible pair. Appendix A gives the derivation and verifies every step against the sample at machine precision; the recovered insider share ranges from 0.4 to 7.2 percent, so these are genuine Shin

solutions rather than a degenerate case. The two rules separate again with three or more outcomes.

Vig-removal rule	n	Gap (pp)	z	p	Break-even (pp)	Verdict
None, raw probabilities	8,014	-2.50	-5.90	<.001	—	artifact
Proportional (primary)	6,840	+0.58	1.22	.223	3.01	below
Additive, equal margin	7,120	-0.60	-1.32	.186	1.88	below
Shin (1993)	7,120	-0.60	-1.32	.186	1.88	below
Constant odds ratio	7,202	-0.74	-1.63	.104	1.79	below
Power	7,211	-1.31	-2.91	.004	1.24	exceeds

Table 5. The primary test under five vig-removal rules, plus the no-normalization case. Verdict compares the absolute gap with that rule's own break-even requirement. Shin and additive coincide exactly for a two-outcome book.

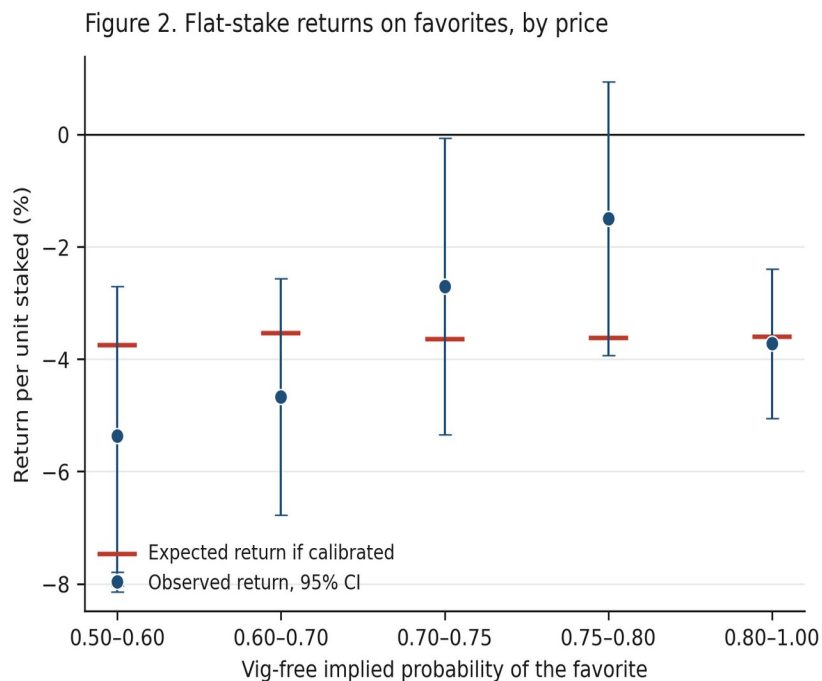
The result is that the primary conclusion is robust to four of five normalization rules examined, but power normalization produces a statistically significant favorite bias that marginally exceeds its own estimated break-even threshold. Four rules leave the gap below their own threshold and none of those four is significant at .05. Power normalization gives -1.31 points against a 1.24-point threshold at $p = .004$: a detectable favorite bias, in the direction of betting underdogs, clearing the economic bar by seven hundredths of a percentage point.

Two reasons are offered for regarding that specification as the least persuasive of the five, and neither is that it produced an inconvenient answer. First, raising probabilities to a common exponent is a flexible transformation without the same direct behavioral or structural interpretation as the others: proportional preserves relative pricing, additive imposes an equal margin, and Shin follows from a model of insider money. Second, it departs furthest from the other rules where the sample is thinnest, pushing heavy favorites to higher implied probabilities than any other rule and thereby concentrating much of the estimated gap in the region with the fewest observations. Neither reason is decisive. A reader who prefers power normalization should read this sample as showing a small, marginally exploitable favorite bias, and that reading is legitimate on the evidence presented here.

5.5 Returns

The dollar-return results tell the same story. Flat one-dollar bets at the quoted prices lose money in every direction: -4.06 percent per bet on all favorites (95% CI: -5.13 to -3.00), -3.91 percent on all underdogs (-6.57 to -1.25), and -2.90 percent on favorites

above .70 (-4.04 to -1.76). Those losses are consistent with the 3.77 percent mean overround. Figure 2 shows that observed returns in every bucket are within the confidence interval around the return expected when prices are calibrated. Put directly against the economic hurdle, the observed +0.58-point calibration gap, and even its +1.26-point upper bound, fall well short of the +3.01 points a favorite better needed at these prices.

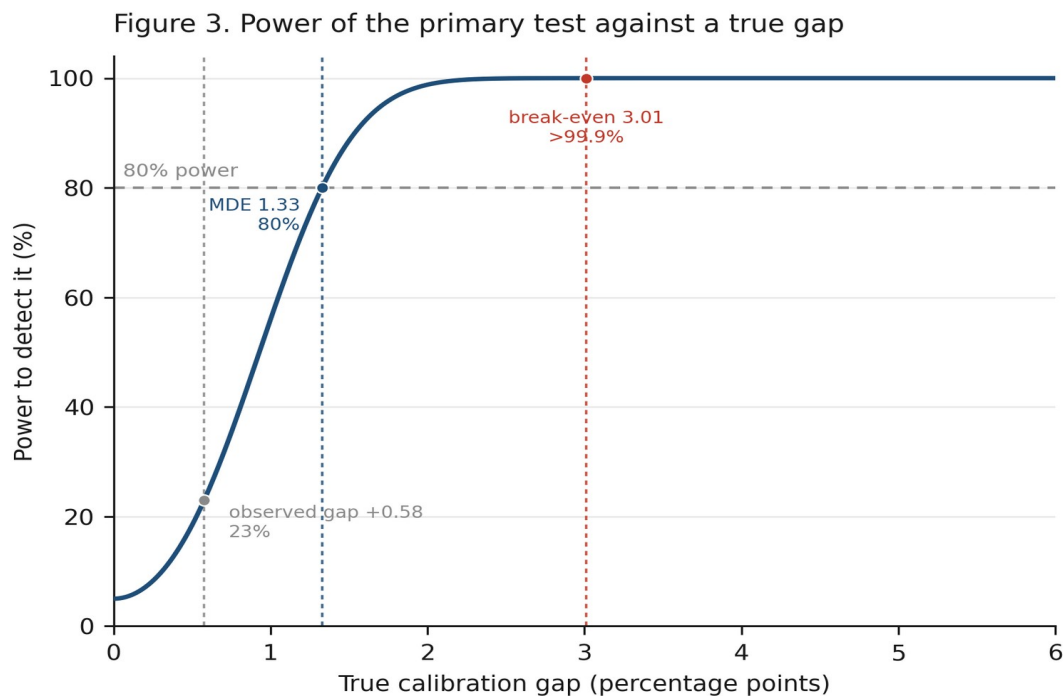


Source: consensus NBA moneylines, 2007-08 to 2019-20, 15,351 regular season games. Vig removed proportionally. Flat one-unit stakes at the quoted price.

Figure 2. Flat one-dollar stakes on the favorite at the quoted price, by bucket of vig-free implied probability, with 95 percent intervals. The expected-return marker in each bucket is computed from the actual quoted odds in that bucket under the assumption that the vig-free probabilities are correct; it differs across buckets because the payout structure differs. Archived consensus moneylines, 2007-08 to 2019-20.

5.6 A methodological caution

Testing observed win rates against raw implied probabilities, without removing the vig, produces an apparent favorite bias of -2.50 percentage points with $z = -5.90$. Under proportional vig removal that gap is the bookmaker's margin rather than anything about better behavior. The artifact reading is conditional on the normalization being approximately correct, and both raw and normalized results are reported so a reader can see exactly what the assumption is doing. The point matters because several bucket-test designs in circulation skip vig removal entirely, and Table 5 shows that the sign of the headline finding flips with that single step.



Two-sided test at $\alpha = 0.05$, $n = 6,840$ favorites above .70 vig-free implied probability. Standard error 0.474 pp from the heterogeneous per-game null variance. All three marks are computed from this sample. The 80% MDE shown is the normal approximation, 1.3281 pp; the exact Poisson-binomial inversion gives 1.3242 pp.

Figure 3. Power of the primary test against a true calibration gap, two-sided at $\alpha = .05$, $n = 6,840$, standard error 0.474 pp from the heterogeneous per-game null variance. Every mark is computed from this sample: the observed +0.58-point gap, the 1.33-point minimum detectable effect, and the 3.01-point break-even requirement. No mark is taken from outside the dataset; see the note in section 6 on why.

6. Discussion

The null result needs to be read narrowly. In these archived consensus moneylines, pooled across 2007 to 2020 and within every season and pre-specified odds range examined, the aggregate favorite calibration gap was smaller than the design's 1.33-point minimum detectable effect and well below the 3.01-point break-even requirement. The estimate holds up under variance estimators that allow correlation within seasons and within teams.

What the result does not establish is broader. It does not rule out localized situational biases outside the pre-specified slices, mispricing at individual books, inefficiency in opening lines, or edges available by shopping for the best price across books. It says nothing about informational efficiency in general, and nothing about the post-2020 market. Because the timing of each quote is undocumented, it also cannot speak to closing-line efficiency, and this paper deliberately makes no claim about how professional bettors benchmark themselves.

The finding is consistent with the broader literature. Moskowitz (2021) reaches a similar conclusion for betting markets generally: whatever behavioral patterns exist in prices are too small to survive costs. The contribution here is the form of the test rather than the surprise of the answer. One pre-specified comparison, a power justification fixed in advance, an economic anchor computed from the actual quoted prices, a data layer verified against an independent extraction, and a robustness program that reports the one specification which does not agree.

The design's power is what makes the null meaningful, and Figure 3 states the claim in the only units this paper can defend. Power exceeds 99.9 percent at the 3.01-point break-even requirement, so a bias large enough to be exploitable at these prices would have been detected almost regardless of where in the range it sat. Power is 80 percent at 1.33 points, well below that threshold. A narrow bucket of a few dozen games has no comparable claim: at 74 games the same calculation gives about 11 percent.

Read together with Table 5, the more durable contribution is methodological, and it is larger than the vig artifact alone. Three treatments of the same 15,351 games give an estimated favorite bias of -2.50 points at $p < .001$ with raw probabilities, +0.58 points at $p = .223$ under proportional normalization, and -1.31 points at $p = .004$ under power normalization. The existence, the magnitude and the sign of the estimate all move with a modeling choice that most papers in this literature make in a single sentence and never revisit.

That is the finding worth carrying forward. An analyst who compares outcomes with raw implied probabilities here finds a 2.5-point favorite bias and would reasonably report it as behavioral; it is the bookmaker's margin. An analyst who picks a normalization rule without stating why can land on either side of zero. A pre-specified design with an economic anchor and a power justification does not make the choice for you, but it makes the consequences of the choice visible, and it is the part of this design most worth reusing.

One thing Figure 3 deliberately does not do is mark effect sizes drawn from prior work. An earlier draft marked 1.7 and 5.1 percentage points as the smallest and largest favorite-bias effects reported in the cited literature. Those two numbers could not be traced to any source in that citation set, and the attempt to source them turned up a substantive problem rather than a clerical one. Six of the eight cited papers report rates of return or point-spread cover rates, not win-probability calibration gaps, and the two quantities are not interchangeable without the odds attached. The one literature review among them, Newall and Cortis (2021), is

entirely directional and reports no effect magnitudes. Converting the classic return estimates into calibration gaps at these prices spans a far wider range than 1.7 to 5.1 and points in the opposite direction, because that work concerns parimutuel horse racing. The marks were therefore removed rather than re-sourced, and the power argument is made against this sample's own break-even threshold instead. A related caution applies to the moneyline branch of the literature generally: Woodland and Woodland (2001) is the closest antecedent, its conversion method has been questioned by Berkowitz, Depken and Gandar (2018), and the authors themselves reported in 2011 that the effect disappears in later seasons.

7. Limitations

Several limits define the reach of these findings. The sample includes regular-season games only and ends in March 2020, where the source archive ends, so the conclusions describe that period. The archive provides one consensus line per game, which means the data cannot speak to line shopping, best-price execution, or the identity of the sportsbook, and the timing of each quote is undocumented at the source.

Proportional vig removal is one of several possible normalizations. Four alternatives are reported in section 5.4; four of five leave the conclusion unchanged and power normalization does not, which is stated rather than set aside. The archive has no ticket-share data, so claims about public betting cannot be tested here. The 260-game primary-source audit was drawn by hand rather than by a seeded random procedure and should be read accordingly; the full-sample cross-validation is the stronger evidence.

The primary test assumes games are independent conditional on their prices. Cluster-robust estimators and block bootstraps relax that assumption and leave the primary result unchanged, but the season-level estimators rest on only thirteen clusters, which is too few for cluster-robust variance to be reliable, and the two-way estimate of the calibration regression should be treated with corresponding caution in both directions. Equal-weight calibration is also a different question from dollar-weighted profitability, which is why the flat-stake return analysis is reported directly.

Finally, the analysis plan was fixed before outcome data were examined, but the record of that freeze is internal to the author's files rather than lodged with an outside registry, so its timing is not independently verifiable and the work is described as pre-specified rather than pre-registered. The plan itself is published in

full with the replication materials, and the design leaves little room for undisclosed flexibility: there is one named primary comparison, a threshold and an effect-size floor both set from stated reasoning rather than from the outcome data, and exploratory analyses reported separately and labeled as such. Readers who wish to check the stronger claim can re-run the confirmatory pipeline, which reproduces every reported statistic from the raw archive.

8. Conclusion

A pre-specified, economically anchored test, adequately powered for its pre-specified primary estimand and run on cross-validated archival data, finds the archived NBA consensus moneylines in this series calibrated: over 15,351 games, favorites were priced correctly to within a fraction of the bookmaker's own margin, and the estimate holds under variance estimators that allow correlation within seasons and within teams.

For bettors, the practical lesson is that in the archived consensus prices examined here, simply choosing favorites or underdogs did not offer an exploitable edge. That statement is deliberately confined to this series and this period; it is not a claim about the NBA betting market in general, and the one normalization rule that disagrees is reported in section 5.4.

For researchers, the sharper conclusion is about method. A pre-specified, economically anchored calibration test under proportional normalization finds no meaningful aggregate favorite bias in this series, and that finding survives correction for dependence. But the estimated existence and even the direction of favorite bias here depend materially on how the bookmaker's margin is removed, and that dependence is invisible to the underpowered single-specification bucket tests this literature has relied on. Reporting the normalization sensitivity is not a caveat attached to the result. On this evidence it is the result.

Data and Code Availability

Data. The underlying historical odds archive is publicly available from sportsbookreviewsonline.com and is redistributed in the repository accompanying Dotan (2020) at <https://github.com/guydotan/ucla-thesis>. The derived analysis dataset is published with the replication materials, together with a script that re-downloads the complete upstream files, prints their SHA-256 hashes, and confirms that every retained column matches the original row for row.

Code. All scripts required to reproduce the reported results, tables and figures are available at <https://github.com/jacobburdier05/nba-moneyline-calibration>. A single command runs verification, dataset construction, the confirmatory analyses, the robustness analyses, the figures and the Appendix A verification in about three minutes.

Analysis plan. The pre-specified analysis plan is published in the same repository under preregistration/analysis_plan.md, with a provenance note stating when it was transcribed into the repository and what evidence does and does not exist for the date it was frozen. It fixes proportional normalization as the primary rule, which is why that choice is not a post hoc selection.

Appendix A. Shin's normalization equals the additive rule for a two-outcome book

Section 5.4 reports that Shin (1993) and the additive rule give identical results in Table 5. Because that claim is unusual and it affects how Table 5 should be read, the derivation is given here rather than asserted. Every step is verified against the sample by src/shin_equivalence_proof.py in the replication materials.

Let q_1 and q_2 be the raw implied probabilities of the two sides and $P = q_1 + q_2$ the booksum, with P greater than one. Shin sets

$$pi_i = [\sqrt{ z^2 + 4(1 - z) q_i^2 / P } - z] / (2(1 - z))$$

with z chosen so the pair sums to one. The additive rule sets $pi_i = q_i - (P - 1)/2$.

Step 1. Invert for z

Rearrange to $2 pi_i (1 - z) + z = \sqrt{ z^2 + 4(1 - z) q_i^2 / P }$ and square both sides. The z -squared terms cancel, and dividing through by $4(1 - z)$, which is nonzero for z below one, leaves

$$pi_i^2 + z pi_i (1 - pi_i) = q_i^2 / P, \quad \text{so} \quad z = (q_i^2 / P - pi_i^2) / (pi_i (1 - pi_i))$$

Step 2. Require both sides to return the same z

Shin's z is a single property of the book, so the expression above must give the same value for each side. Because the pair sums to one, $1 - pi_1$ equals pi_2 and $1 - pi_2$ equals pi_1 , so both denominators equal $pi_1 pi_2$ and the requirement reduces to equal numerators:

$$(q_1^2 - q_2^2) / P = pi_1^2 - pi_2^2, \quad \text{that is} \quad (q_1 - q_2)(q_1 + q_2)/P = (pi_1 - pi_2)(pi_1 + pi_2)$$

Since $q_1 + q_2 = P$ by definition and $p_{i_1} + p_{i_2} = 1$ by the constraint, this collapses to $q_1 - q_2 = p_{i_1} - p_{i_2}$. Shin's rule preserves the difference of the raw probabilities.

Step 3. Solve

Two linear conditions now pin the pair down uniquely: the sum is one and the difference is $q_1 - q_2$. Therefore $p_{i_1} = (1 + q_1 - q_2)/2$. The additive rule gives $q_1 - (q_1 + q_2 - 1)/2$, which is the same expression. The two rules coincide.

Numerical verification and scope

Across the 15,350 games in the sample with a positive margin, the maximum absolute error is $3.3e-16$ for the difference-preserving property in step 2, $2.8e-16$ for the closed form in step 3, and $2.8e-16$ for the equivalence itself, all at the limit of double precision. The recovered insider share z is consistent across the two sides to $1.5e-14$ and ranges from 0.0037 to 0.0718 with a mean of 0.0384, so these are genuine interior Shin solutions and not a degenerate corner. One game in the archive is quoted with a booksum below one, where Shin's model has no valid solution; it is excluded from this appendix and falls back to proportional in Table 5.

The equivalence does not mean the two rules express the same idea. Shin's is derived from a bookmaker pricing against insider money and is intended to load margin onto longshots; the additive rule is a mechanical convention with no behavioral content. They coincide only because a two-outcome market leaves a single degree of freedom once the sum constraint and the difference-preserving property are both imposed. With three or more outcomes they separate. The practical consequence for this paper is narrow: a reader who asks why Shin was not tested should know that it was, and that in this market it is the additive row.

One implementation note, recorded because it produced wrong numbers before it was caught. Evaluated directly, the numerator $\sqrt{z^2 + a} - z$ is a difference of nearly equal quantities as z approaches one, and in double precision it loses most of its significant digits, which sends the root finder to values whose probabilities sum to roughly 0.98 rather than one. Multiplying by the conjugate gives the algebraically identical and numerically stable form $p_i = 2 z^2 / [P (\sqrt{z^2 + a} + z)]$, which is what the code uses.

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Reproducibility and AI Disclosure

This paper is designed to be reproducible. The replication materials include the frozen analysis plan, the primary and robustness analysis scripts, the data verification script and its full output, the figure-generating scripts, and machine-readable result tables, so every number and figure can be regenerated from the raw archive. The headline statistics were reproduced independently from the raw files, including the exact test size and power, the Poisson-binomial test statistic, the break-even algebra, the Cochran Q construction, and the alternative normalizations.

AI tools assisted with code generation, statistical checking, data verification, and copy-editing. The research question, study design, assumptions, interpretation, and final claims are the author's. No AI-generated source or statistic is cited as evidence, and every reference in this paper was located and read directly.